

Proactive Cooking Assistance via Dual-Tier Retrieval-Augmented Generation on Egocentric XR Video

PIC2 - Master in Computer Science and Engineering
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Abstract As multimodal artificial intelligence and extended reality continue to advance, their convergence in procedural task assistance introduces critical new challenges. Cooking is a paradigmatic case: users engage both hands continuously, manage concurrent steps under time pressure, and cannot pause to query an AI assistant mid-task. Effective cooking assistance must therefore be entirely proactive, with the agent autonomously deciding when and what to communicate based solely on continuous visual observation. While prior research has produced retrieval-augmented generation (RAG) systems for recipe knowledge and egocentric video understanding benchmarks, two fundamental gaps remain: how an agent should decide autonomously when to retrieve from a knowledge source and issue an unprompted safety alert or step correction (Proactive RAG Activation), and how to fuse static recipe domain knowledge with live egocentric object-state perception in real time (dual-tier knowledge retrieval). To address these gaps, this work proposes and evaluates a proactive multimodal cooking assistant built around a dual-tier memory architecture that separates a static recipe knowledge graph ($\mathcal{K}_1$) from a live object-state buffer ($\mathcal{K}_2$), and an endogenous intervention policy that fires without any user query. A mixed-design user study is planned using a scrambled-egg cooking task in a controlled laboratory kitchen: memory tier is manipulated between subjects and intervention mode (reactive versus proactive) within subjects, with each participant cooking twice. The evaluation tests three pre-registered hypotheses concerning interruption-timing accuracy, procedural error rate, perceived intrusiveness, and guidance relevance. This work aims to contribute both to the scientific understanding of proactive intervention in time-critical procedural tasks and to the practical engineering of dual-tier RAG architectures for real-world, hands-free cooking assistance.

Keywords — Extended Reality, Egocentric Video, Retrieval-Augmented Generation, Proactive AI Agent, Cooking Assistance

*I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa (<https://nape.tecnico.ulisboa.pt/en/apoio-ao-estudante/documentos-importantes/regulamentos-da-universidade-de-lisboa/>).

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1 Introduction

The maturation of extended reality (XR) headsets and the rapid advancement of multimodal large language models have together created a new class of intelligent systems capable of perceiving, reasoning about, and responding to the physical world in real time [1]. Wearable egocentric devices now provide continuous first-person video streams that capture the user’s hands, tools, and immediate environment without requiring deliberate interaction. Recent advances in efficient multimodal architectures have further demonstrated that low-latency visual-language inference can be achieved on consumer-grade and localized hardware [2], a prerequisite for XR-embedded assistants that cannot tolerate the round-trip latency of cloud inference during time-critical tasks. In parallel, retrieval-augmented generation (RAG) has established itself as the dominant paradigm for grounding model outputs in structured domain knowledge, substantially reducing hallucination and improving factual precision in video and text-based question-answering systems [1, 3]. Together, these developments point toward a compelling near-term application: an AI agent that watches a person work and offers expert guidance at exactly the moment it is needed, without ever being asked.

Cooking is a uniquely demanding domain for such a system. It is a time-critical procedural activity that requires simultaneous management of temperature, ingredient state, sequential steps, and timing, frequently under conditions where the user’s hands and visual attention are fully occupied. Prior work in AR-guided procedural support has demonstrated that novice practitioners experience significant cognitive overload when attempting to follow digital instructions while physically performing a task [4, 5]. Timing failures, missed steps, and safety incidents occur precisely at the moments when consulting an external system is least feasible. A cooking assistant that requires the user to pause and ask a question does not solve this problem; it reproduces the exact bottleneck it was intended to eliminate.

Existing research has addressed parts of this challenge in isolation. RAG architectures for recipe knowledge, such as RecipeRAG [3], have shown that structured culinary information can be retrieved accurately given a textual query. Egocentric video benchmarks, including LifeEval [6] and Minerva-Ego [7], have shown that state-of-the-art multimodal models can reason over first-person video when prompted to do so. Proactive agent frameworks such as YETI [8] and SensibleAgent [9] have further demonstrated that an agent can detect intervention opportunities from continuous egocentric streams without an explicit user trigger. However, these contributions have not been unified into a system that autonomously decides *when* and *what* to communicate during a live cooking task. Critically, video RAG systems such as StreamRAG [1] are evaluated in streaming question-answering settings where a user still formulates and issues a query, an interaction model that is architecturally incompatible with a hands-busy physical task.

Two interdependent problems thus remain open. The first is the *Proactive RAG Activation* problem: how should an agent decide, based solely on continuous visual observation, when it is necessary to retrieve from a cooking knowledge source and issue an unprompted intervention, whether a safety alert such as “the oil is smoking, reduce the heat” or a technique correction? The second is the *dual-tier knowledge retrieval* problem: a cooking agent must fuse internal static domain knowledge (for example, safe temperature thresholds or correct ingredient ratios from a recipe source) with external dynamic evidence drawn from live object-state perception [10]. Current video RAG models retrieve exclusively from video history; they have no mechanism for cross-referencing live egocentric observations with a structured recipe knowledge base in real time. The Evidence-Decision-Feedback framework [11] argues that effective adaptive agents must separate evidence extraction, decision-making, and response generation as distinct architectural concerns, yet no cooking assistant has implemented this separation with egocentric video as the evidence source. The Centaur Evaluations framework [12] further argues that AI systems assisting humans should be evaluated on the quality of human augmentation rather than autonomous task performance, a criterion that no existing cooking benchmark captures.

This thesis proposes and evaluates a proactive multimodal cooking assistant that continuously streams egocentric XR video, monitors the culinary scene using object-centric state tracking, and delivers unprompted guidance by retrieving from both a static cooking knowledge source and a dynamic video history index. The experimental evaluation is intentionally scoped to a single standardized benchmark recipe, a deliberate methodological choice that eliminates performance variance introduced by recipe

complexity and ingredient diversity, enabling clean isolation and measurement of the agent’s core architectural variables.

1.1 Research Questions

This work is guided by three research questions, each corresponding to a distinct gap identified in the literature:

- **RQ1:** To what extent does a proactive intervention policy improve interruption-timing accuracy and reduce procedural error rate during a hands-busy cooking task, compared to a reactive, query-driven baseline?
- **RQ2:** Does a well-timed proactive policy increase, decrease, or leave unchanged the perceived intrusiveness of assistance relative to a system that responds only when explicitly queried?
- **RQ3:** To what extent does a dual-tier RAG architecture, fusing static recipe knowledge with a live object-state buffer, yield higher guidance relevance than a single-tier baseline that retrieves from either source alone?

1.2 Work Objectives

The primary objective of this work is to design, implement, and evaluate a proactive multimodal cooking agent that operates without user prompting. This requires four sequential objectives: (1) constructing a streaming egocentric video pipeline for XR devices capable of low-latency frame ingestion and event segmentation; (2) implementing an object-centric state monitoring module that detects culinary events such as ingredient state transitions and temperature cues in real time; (3) designing a dual-tier RAG system that separates static recipe domain knowledge ($\mathcal{K}_1$) from a live object-state buffer ($\mathcal{K}_2$) and fuses both tiers at inference time; and (4) validating the full system through a controlled mixed-design user study in which participants are assigned between-subjects to a memory-tier group and experience both intervention modes within-subjects, cooking scrambled eggs twice per session.

1.3 Expected Contributions

This thesis makes three contributions:

1. A proactive intervention architecture for real-time egocentric cooking assistance that continuously monitors the culinary scene and issues unprompted suggestions and alerts without requiring user-initiated queries.
2. A dual-tier RAG design that separates static recipe domain knowledge ($\mathcal{K}_1$) from a dynamic live object-state buffer ($\mathcal{K}_2$) and fuses both retrieval tiers at inference time to produce contextually grounded, state-aware suggestions.
3. An empirical evaluation through a mixed-design user study providing the first systematic comparison of proactive versus reactive cooking assistance and single-tier versus dual-tier retrieval under controlled conditions, yielding evidence evaluating three pre-registered hypotheses.

2 Background

The proactive intervention architecture and dual-tier retrieval design introduced above rest on a set of established technical concepts that must be made precise before the system can be specified or evaluated. The retrieval and memory structures, the egocentric streaming pipeline, and the human-factors constructs are defined here; concrete systems and their relative merits are treated in the Related Work.

2.1 Retrieval-Augmented Generation and Dual-Tier Memory

Retrieval-Augmented Generation (RAG) couples a generative model with an external knowledge base that is queried at inference time. A baseline RAG pipeline maps an input signal into an embedding space, retrieves the most similar entries from a knowledge store, injects them into the model context, and conditions generation on the retrieved evidence [13]. RAG therefore functions as an external memory that offloads knowledge from the model parameters and grounds outputs in an inspectable source.

Hierarchical retrieval-augmented architectures distinguish between two distinct knowledge surfaces. The static Tier-1 store $\mathcal{K}_1$ is a persistent, slow-changing structured representation of domain knowledge: a cooking procedure expressed as a directed acyclic graph whose nodes are ingredients and actions and whose edges encode pre-conditions, post-conditions, and timing constraints [3, 14]. The dynamic Tier-2 store $\mathcal{K}_2$ is a high-velocity buffer of time-indexed, object-centric trajectories that record the unary property transitions of tracked entities (for example, an egg progressing from liquid to coagulating to solid) [10, 15]. A single-tier architecture exposes one homogeneous retrieval surface; a dual-tier architecture maintains both $\mathcal{K}_1$ and $\mathcal{K}_2$ at distinct update rates and fuses them at inference time [16, 17]. The canonical monitoring primitive is the State–Event–State transition $S_{t-1} \xrightarrow{E_t} S_t$, where $\mathcal{K}_1$ encodes the expected transitions and $\mathcal{K}_2$ records the observed ones [18].

Two execution triggers follow from this structure. A reactive trigger fires retrieval only on an explicit user prompt u_t , conditioning generation on $\text{Retrieve}(u_t, \mathcal{K})$; this is the standard reactive RAG baseline. A proactive trigger fires retrieval on an endogenous state predicate $\phi(s_t, \mathcal{K}_1, \mathcal{K}_2)$ with no user prompt, governed by a severity decision variable $D_t \in \{0, 1, 2\}$ that gates whether to retrieve and communicate, and at what urgency level [8, 19].

2.2 Egocentric XR Streaming Pipeline

An egocentric stream is a head-mounted, first-person video feed characterised by rapid field-of-view shifts, hand occlusions, motion blur, and a non-stationary workspace, which make it structurally distinct from a fixed third-person camera feed [7]. The closed-loop assistance pipeline assumed here is device-agnostic and comprises five blocks: (i) capture of frames from the headset camera at a configurable rate; (ii) transport of frame packets over a low-latency channel to a processing node; (iii) perception, an object-state extractor that updates $\mathcal{K}_2$; (iv) reasoning, the fusion of $\mathcal{K}_1$ and $\mathcal{K}_2$ feeding the intervention policy; and (v) feedback, the low-latency response returned to the user. Closed-loop assistance requires the end-to-end latency to remain well below the task-critical transition time, and bounded sliding windows or event segmentation keep token and memory growth finite over an unbounded stream [1, 18]; the live knowledge tier $\mathcal{K}_2$ described in Section 4.7 is the structural component that enforces this boundedness constraint in the proposed architecture.

2.3 Proactive Assistance and Study Variables

Proactivity in automation denotes the transition of an agent from an obedient tool, which acts only on an explicit prompt u_t , to an autonomous collaborator that initiates communication when ϕ holds; it is grounded in the Just-in-Time Adaptive Intervention mapping of a decision point and tailoring variables to an intervention option [19].

Three outcome constructs follow from this framing. Guidance relevance is the semantic match between a delivered suggestion and both the current $\mathcal{K}_2$ state and the applicable $\mathcal{K}_1$ rule. Interruption timing accuracy is the temporal alignment between the intervention time t_{int} and the optimal point within a recipe-critical decision window. Perceived intrusiveness is the degree to which an unprompted alert displaces primary-task attention; its underlying model is the split-attention and working-memory cost incurred when a user manages a physical task and an AI feedback channel simultaneously [20]. Following the centaur framing, overall success is defined as joint human–system performance rather than autonomous task completion alone [12].

3 Related Work

Our proactive, dual-tier cooking assistant proposed in this work operates at the convergence of three distinct research paradigms: egocentric video understanding, intelligent cooking and guidance systems, and multimodal retrieval-augmented generation. Rather than surveying these areas chronologically, this section organises the literature into three thematic buckets and critiques each one against the constraints that define a real kitchen: a user whose hands and visual attention are fully occupied, an unbounded streaming video feed, high cognitive load, and state transitions that become irreversible within seconds. Each subsection exposes a structural limitation of the current state-of-the-art and uses that limitation to motivate one of the hypotheses tested in this work.

3.1 Egocentric Video Understanding and Procedural Tasks

First-person action recognition has matured rapidly on the back of large-scale egocentric corpora. Datasets such as Ego4D [21] and EPIC-KITCHENS [22] established the modern baselines for tracking objects, hands, and step progression in kitchen activity, and they remain the reference benchmarks for the field. Their defining characteristic, however, is also their fundamental limitation for the present application: they are curated as offline corpora to be analysed retrospectively, not as substrates for live, streaming inference under a latency budget. Models trained and evaluated on them answer questions about recorded video rather than committing to actions while the video is still unfolding.

This offline, passive orientation persists in the most recent egocentric reasoning systems. Minerva-Ego shows that even state-of-the-art vision-language models fail on the perceptual primitives that cooking demands, namely object identification during manipulation and temporal grounding, and that the proposed remedy is post-hoc visual prompt engineering rather than an agent that determines its own spatial and temporal focus on the fly [7]. LifeEval confirms the pattern at the level of assistive interaction: although it models real-time scenarios, it evaluates assistants through a query-triggered protocol, top proprietary models degrade sharply when moving from recognition to dynamic task reasoning and goal-oriented planning, and its safety-and-feasibility dimension is assessed only in response to a user-initiated question [6]. In other words, the benchmark itself never asks the model to intervene unprompted.

A second weakness concerns the semantics of what is perceived. Current networks largely label actions such as “stirring” or “pouring” without reasoning over the downstream pre- and post-conditions encoded in a recipe. This is acute for cooking, where the salient event is an intra-object state change: an egg progressing from liquid to coagulating to solid is a non-uniform, spatially progressing transformation that frame-level, uniform-state perception cannot capture, as the SPOC formulation demonstrates [15]. Efficient on-device perception removes the deployment barrier, since compact multimodal models can now recognise events such as overheating in real time on resource-constrained hardware [2], yet recognition capability alone carries no temporal policy for when to act. Finally, work in food computing remains anchored to static, final-outcome imagery and text mappings, explicitly lacking ingestion of live egocentric streams and modelling of intermediate cooking states [23].

Taken together, the egocentric literature provides an increasingly capable perception substrate that is nonetheless offline, passive, query-triggered, and semantically shallow with respect to recipe rules. A system intended to help a hands-busy cook must therefore add two faculties the field currently lacks: the autonomy to decide *when* to speak, and a structured memory of the evolving physical state. These two requirements motivate the proactivity argument of the next subsection and the memory argument of the last.

3.2 AI Cooking Assistants and Intelligent Guidance Systems

Existing digital cooking assistants are overwhelmingly reactive. Conversational recipe and dietary systems such as DietQA [24] and knowledge-graph-driven generators such as RecipeRAG [3] excel at parsing complex, multi-constraint requests, but they require the user to formulate and issue a textual query. This interaction model is architecturally incompatible with a kitchen in which hands are messy and food is already on the heat: the moment the system is most needed is precisely the moment the user cannot

stop to type or phrase a question. Augmented-reality guidance partially addresses the input modality, in that head-mounted displays keep the hands free and outperform manuals and tablets for spatial cooking tasks [4], yet such guidance is typically delivered as static, step-by-step instruction that can cultivate mindless instruction-following rather than understanding [5].

The cost of forcing the user to drive the interaction is not merely inconvenience. Cognitive-ergonomics analysis shows that splitting attention between physical manipulation and an AI feedback channel inflates extraneous load, degrading both task and monitoring performance [20], and the Just-in-Time Adaptive Intervention paradigm argues that effective help must be bound to a decision point in the task rather than to an explicit prompt [19]. A purely reactive interface thus reproduces, at the level of cognitive friction, exactly the bottleneck it was meant to remove.

A smaller body of work has begun to make agents proactive. YETI introduces an ultra-lightweight gate that maps a continuous egocentric feed to a binary intervene-or-wait decision, but it deliberately omits any semantic recipe checklist or state machine tied to a master plan [8]. Sensible Agent optimises *what* assistance to offer and *how* to deliver it in AR, reducing interaction effort, yet it is not designed for high-stakes settings where physical state mutates continuously [9]. Em-Garde demonstrates that proactive streaming understanding can run at a stable frame rate on a single GPU by decoupling semantic reasoning from lightweight streaming perception [25], establishing feasibility on wearable-class hardware. Crucially, however, the proactive dialogue system ProAssist reports that real-world execution falls below acceptable human-rated thresholds specifically because of poor response timing [26], identifying *when to speak* as the open problem rather than *what to say*. This is reinforced by the human-AI collaboration literature, which frames interruption as a friction-laden act and argues that such systems should be judged on joint human-system performance and on the tension between task performance and behavioural comfort [11, 12, 27].

The state-of-the-art therefore presents a clean dichotomy: assistants are either reactive, and so impose a query bottleneck in hands-busy cooking, or proactive, but with unreliable timing, no recipe-grounded state, and intrusiveness that has not been measured in a time-critical culinary task. This gap leads directly to the first two hypotheses.

H_1 : Replacing a reactive, query-driven interface with a proactive intervention policy improves interruption-timing accuracy and lowers the procedural error rate during a hands-busy cooking task.

H_2 : A well-timed proactive policy does not increase, and may reduce, perceived intrusiveness relative to the cognitive cost of user-initiated querying, despite delivering unprompted alerts.

3.3 Multimodal RAG and Knowledge Fusion Architectures

Retrieval-augmented generation has become the dominant strategy for grounding model outputs in external knowledge, and recent surveys position the integration of large language models with structured knowledge along a spectrum: a single homogeneous text-retrieval surface occupies the lower levels, while context-aware mechanisms that track environmental state represent the upper level of optimisation [13]. The same surveys flag the time overhead of constructing and updating knowledge in real time as an unresolved obstacle, which is exactly the regime a streaming cooking assistant must operate in.

The first failure mode of current systems concerns *where* knowledge is stored. A large family of culinary systems retrieves from a single static store. Action-centric ontologies represent recipes as directed acyclic graphs of processes, transfers, and state changes [14]; knowledge-graph recipe generators [3], context-aware knowledge-graph messaging pipelines [28], food substitution graphs and surveys [29, 30], and commonsense substitution frameworks [31] all encode rich, structured culinary knowledge. Yet they are uniformly static: their predicates remain symbolic, with no architecture for verifying them against live sensory evidence, so a system built on them alone cannot react when an ingredient is dropped, burned, or already past the target state.

The second failure mode concerns systems that track live visual state but treat it as a single tier. Streaming and video RAG approaches such as Event-Causal RAG segment a stream into State-Event-State structures to avoid temporal fragmentation [18], and object-centric monitoring such as Tracking the

Truth builds structured historical trajectories that markedly improve consistency [10]; related streaming pipelines achieve the necessary low latency through event segmentation and token reuse [1]. These systems, however, are evaluated as passive, post-hoc viewers answering questions about a stream, and they retrieve from video history alone, with no mechanism for cross-referencing observations against a structured recipe rule base. The complementary necessity of separating fast-changing state from slow archival knowledge is established outside the cooking domain by versioned hot/cold vector architectures [17] and by dual-memory agents that pair an internal trajectory memory with an external knowledge store [32], both of which improve streaming efficiency and decision robustness but neither of which is grounded in egocentric culinary perception.

The picture that emerges is that no existing system fuses the two knowledge surfaces a cook actually needs. Static recipe knowledge bases cannot track the live transition of ingredients, while live video indices have no access to the recipe rules that make an observed transition meaningful. A single-tier baseline, drawing on either the recipe text or the latest visual frame, must therefore produce help of degraded relevance precisely when the physical state diverges from the plan. This motivates the final hypothesis.

H₃: A dual-tier memory architecture, fusing static recipe rules with a live object-state buffer, yields higher guidance relevance than a single-tier baseline that retrieves from either source alone.

4 Solution

The three gaps identified in Sections 3.1–3.3 demand a concrete engineering response: an agent that watches a cooking task continuously, knows what the recipe expects, tracks what the food is actually doing, and acts before failure becomes irreversible, all without any prompt from the user. The architecture built to meet these requirements is described here. Four constraints govern every design decision: **closed-loop latency**, meaning the end-to-end delay from frame capture to notification must stay well below the seconds-scale window of a cooking state transition; **bounded memory**, since the egocentric feed is theoretically infinite and must be segmented into clearable windows to prevent linear token growth [1, 18]; **one-way output**, with the assistant delivering compact notifications and never opening a dialogue; and **hardware-agnostic capture**, so the pipeline is decoupled from any specific headset. Every module, data store, and decision variable named below corresponds to a labelled block in Figure 1.

4.1 System Overview

Figure 1 shows the high-level pipeline. Egocentric frames flow from the **Capture** module through a low-latency **Transport** channel to a **Proactive Gate** that discards redundant frames before any expensive inference is triggered. Frames that pass the gate enter the **Perception / Tier-2 Extractor**, which maintains a live object-state buffer K_2 . The **Fusion & Reasoning** module compares K_2 against the static recipe knowledge store K_1 and, when a conflict is detected, invokes a multimodal language model to generate a suggestion. The **Intervention Policy** evaluates the endogenous predicate $\phi(s_t, K_1, K_2)$ and emits the severity variable $D_t \in \{0, 1, 2\}$. When $D_t \geq 1$, a **Notification** is delivered to the user’s XR device.

4.2 Capture Module

An egocentric camera mounted on a head-worn XR device provides a continuous first-person video stream. This feed has properties that distinguish it from fixed third-person cameras: rapid field-of-view shifts, hand occlusions, motion blur, and a non-stationary workspace [7]. Research on frame-density saturation shows that moderate sampling of approximately eight frames per short interval hits a performance ceiling for real-time egocentric assistants, and that denser sampling adds latency without accuracy benefit [6]. The capture rate is therefore set within this range to balance perceptual quality against processing load.

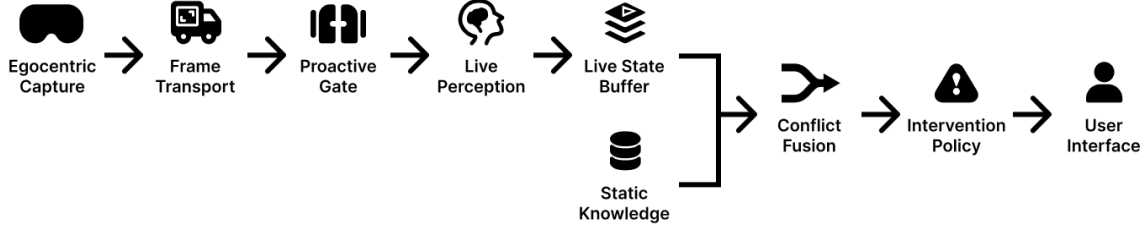


Figure 1: High Level System Architecture Overview

The module exposes a uniform frame-packet interface to the Transport layer regardless of the specific headset vendor.

4.3 Transport Module

Frame packets are forwarded from the XR device to the processing node over a low-latency channel, implemented as WebRTC or WebSocket depending on the deployment environment. Only frames that pass the Proactive Gate are forwarded for heavy processing; frames suppressed by the gate are discarded at source to minimise bandwidth and round-trip delay. The network round-trip is included as a fixed term in the closed-loop latency budget.

4.4 Proactive Gate

The gate is the first and cheapest computational stage. It runs in a continuous loop and screens every incoming frame *before* any model inference is triggered, serving the same role as the binary gating mechanism introduced in YETI [8]. Two signals combine to form the screen:

- **Frame similarity.** The pixel-level Structural Similarity Index (SSIM) between consecutive frames detects meaningful scene change. Low-SSIM frames are judged redundant and suppressed [8].
- **Depth score.** A ViT-based similarity score identifies event boundaries in the stream, analogous to the Stream Event Segmentation depth score of StreamRAG:

$$d_i = \frac{c_l^{\text{ViT}} + c_r^{\text{ViT}} - 2c_i^{\text{ViT}}}{2} \quad (1)$$

Frames at local maxima of d_i mark the onset of a new cooking event (e.g., *pouring*, *stirring*, *overheating*) and are forwarded downstream [1].

Together these two signals form a coarse binary screen that keeps system throughput stable. YETI demonstrates that a 1 FPS SSIM gate achieves recall and F1 competitive with heavy frameworks; Em-Garde shows that decoupling cheap gating from expensive reasoning sustains 10–15 fps on a single GPU because the streaming loop never calls the large model [25].

4.5 Perception and Tier-2 Extractor

The extractor implements the **propose-match** design of Em-Garde [25], separating a one-time heavy reasoning phase from a lightweight continuous matching phase.

Propose phase (offline / startup). Before the cooking session begins, **Gemma 4 E4B**, an open-weight 4 B-parameter multimodal model, parses the static recipe knowledge in K_1 and generates a library of *visual proposals*: structured embeddings that encode the expected perceptual signature of each recipe transition (e.g., “egg mixture loses transparency and firms” is mapped to a concrete visual embedding). Gemma 4 E4B is selected over proprietary API models for three engineering reasons: (i) open weights

allow K_2 object-centric trajectory tokens to be injected directly into the prompt sequence alongside image frame tokens, giving precise control over what context the model receives at each inference call; (ii) pinned-checkpoint inference is fully deterministic and reproducible across runs; (iii) a 4 B-parameter model operating at 4-bit quantisation fits within the memory envelope of a local GPU server without requiring a cloud round-trip on every alert [2].

Match phase (streaming). For each frame that passes the gate, a lightweight embedding comparator checks the frame against the visual-proposal library. For state-label assignment the comparator combines **CLIP** for local semantic matching (e.g., “is this egg liquid or solid?”) with **DINOv2** for global frame context, following the recommendation in SPOC to balance fine-grained local cues with spatial scene structure [15]. Object-centric trajectories are maintained for each tracked entity, encoding unary property transitions (e.g., egg: `liquid` $\rightarrow$ `coagulating` $\rightarrow$ `solid` $\rightarrow$ `scorched`), consistent with the chunk-wise state extraction design of STEMO-Track [10]. The result written to K_2 at each step is a time-stamped record of the form $(entity_id, state_label, confidence, t)$.

4.6 K_1 : Static Knowledge Tier

K_1 is a single, unified knowledge graph stored in Neo4j, built offline before the session begins following the KAG enrichment model of $T_{\text{semantic}} = T_{\text{fact}} \oplus T_{\text{concept}}$ [33]: recipe action nodes (T_{fact}) are augmented at ingestion time with domain-physics safety bounds (T_{concept}), so the Fusion module retrieves both in a single graph traversal rather than two sequential store lookups [17].

Raw recipe text is scraped from cooking blogs and food websites and processed through an LLM-driven ingestion pipeline that strips prose to a **canonical action form**, extracting atomic steps, ingredients, quantities, and timing cues [34]. Rather than encoding a single linear recipe, the pipeline produces a **multi-path directed acyclic graph**: common variations found across sources (e.g., *butter vs. oil*, *low-heat slow scramble vs. high-heat fast scramble*) are preserved as parallel branches with shared pre/post-condition nodes.

Each node is typed as *Process*, *Transfer*, or *Plate* following the action-centric ontology of Kumbhakern et al. [14] and carries four fields: (a) pre-conditions: what must be true before execution; (b) post-conditions: the expected ingredient state after execution; (c) timing constraints: the expected duration or deadline; and (d) a HAS_SAFETY_BOUND typed attribute edge on *Process* nodes, encoding the domain-physics upper bound for that step (e.g., coagulation onset at $\approx 62^\circ\text{C}$, full coagulation at $\approx 70^\circ\text{C}$, Maillard browning onset). If K_2 reports visual proxies consistent with exceeding a bound (colour change, texture shift), the attribute edge activates and can raise $D_t = 2$ regardless of recipe-step position, independently of the recipe conflict score.

Edges encode causal dependencies: branch B cannot activate until the post-condition of its parent node A is verified by K_2 . At runtime the Fusion module selects the branch whose pre-conditions best match the current K_2 state, delivering advice consistent with how the user is actually cooking. K_1 is ingested once and remains immutable during a session, mapping to the cold archival tier in a hot/cold versioned knowledge architecture [17]. At reasoning time, the relevant node subgraph is serialised into structured text and injected into the Gemma 4 E4B prompt alongside the K_2 state [28].

4.7 K_2 : Live Knowledge Tier

K_2 is a bounded, time-indexed in-memory buffer that stores the sequence of $(entity_id, state_label, confidence, t)$ records produced by the Tier-2 Extractor. It functions as the **hot tier**: high-velocity, volatile, and optimised for recency queries [17]. Boundedness is enforced by clearing records once the corresponding recipe step is recognised as complete, preventing linear memory growth over an unbounded stream [18].

The State-Event-State (SES) formalism $S_{t-1} \xrightarrow{E_t} S_t$ [18] is the native storage unit: K_2 holds the most recent chain of SES records for each tracked entity, forming the live “actual trajectory” that the Fusion module compares against the expected trajectory in K_1 . This role is analogous to the internal-memory

tier of the M^2 architecture [32]: a dynamic, compressed representation of recent events used to inform the next decision.

4.8 Fusion and Reasoning Module

At each gate-triggered cycle the module executes a two-step process. First, it queries K_1 for the expected post-condition of the current recipe step and K_2 for the latest observed state of each relevant entity. Second, it computes a **conflict/confidence score** that quantifies the deviation between expectation and observation. A large deviation (for example, K_1 expects `egg: coagulating` while K_2 reports `egg: scorched`) signals a potential safety or quality failure.

Only when the conflict score exceeds a threshold is Gemma 4 E4B invoked to generate a suggestion; the model is not called every frame, preserving the latency budget [8, 25]. The scoring approach is grounded in MultiRAG’s Multi-level Confidence Computing [35]: graph-level structural consistency (from K_1) combined with node-level local accuracy (from K_2) provides a mathematically rigorous basis for the two-tier architecture. The KAG semantic tensor formulation $T_{\text{semantic}} = T_{\text{fact}} \oplus T_{\text{concept}}$ [33] maps the fusion directly: K_1 provides T_{fact} (static recipe nodes and edges) while K_2 dynamically generates runtime concept nodes at inference time.

When the model is invoked, it receives the conflicting K_2 state and the relevant K_1 subgraph serialised as text, and is instructed to produce one warning sentence and one corrective suggestion sentence.

4.9 Proactive Intervention Policy

The intervention policy evaluates the endogenous predicate $\phi(s_t, K_1, K_2)$ and maps it to a two-level severity variable $D_t \in \{0, 1, 2\}$ [8, 19]:

- $D_t = 0$: the observed state is within tolerance of K_1 expectations; no intervention is issued and the gate continues monitoring.
- $D_t = 1$ (**Suggestion**): a minor, reversible deviation is detected. A visual-only overlay is delivered (brief, dismissible, no audio) at a JITAI decision point where correction is beneficial but inaction does not cause immediate failure [9, 11].
- $D_t = 2$ (**Alert**): a major deviation is detected; inaction would cause an irreversible outcome such as burning, timer expiry, or a safety violation. A visual overlay combined with an audio cue is delivered and is not dismissible until the state resolves or the user acknowledges [19].

Severity determination. The conflict score from the Fusion module maps to severity as follows: below a lower threshold τ_1 the score yields $D_t = 0$; between τ_1 and an upper threshold τ_2 it yields $D_t = 1$; above τ_2 or when a physics constraint edge in K_1 is violated it yields $D_t = 2$. Grounding for this two-threshold design comes from MultiRAG’s MCC equations [35].

Cooldown suppression. After any $D_t \geq 1$ event the policy enters a suppression window during which repeat firing is blocked. The window clears only when K_2 confirms the state has moved toward the K_1 expected post-condition, or when the window times out. This prevents notification flooding when a problematic condition persists [8, 9].

Worked example. K_1 encodes the transition S_0 (liquid egg, low heat) $\xrightarrow{E_1} S_1$ (softly coagulated, off heat). K_2 instead records E_1 escalating into E'_1 (overheating, rising colour change) $\rightarrow S'_1$ (scorched solid). The conflict score first crosses τ_1 : $D_t = 1$ fires the suggestion “*Your eggs are cooking faster than expected: consider lowering the heat.*” If the state continues to worsen and crosses τ_2 , $D_t = 2$ fires an alert with audio: “*Your eggs are burning: remove from heat immediately.*” This SES-chain escalation is adapted from the Dual-Sentinel paradigm of EC-RAG [18].

The policy is purely endogenous: it fires without any user query u_t , distinguishing it structurally from reactive RAG baselines [13].

4.10 Notification Delivery

The system delivers one of two notification types depending on D_t :

- **Suggestion** ($D_t = 1$): a compact visual-only overlay. The message is one sentence, dismissible, and positioned unobtrusively in the periphery of the field of view to minimise split-attention cost [9, 20].
- **Alert** ($D_t = 2$): a visual overlay combined with a short audio cue. Non-dismissible until the state resolves or the user acknowledges. Positioned centrally with higher visual salience. Audio-visual co-delivery is appropriate for high-urgency, hands-busy contexts where a visual-only notification may be missed [4, 9].

No user input channel exists in either case: the system never opens a dialogue or waits for a response. After delivery, the cooldown suppression window activates and D_t returns to 0 once resolution is confirmed.

4.11 Compute Topology and Deployment Justification

A naïve topology running full multimodal model inference on every frame on-device is infeasible: a 4 B-parameter model at 4-bit quantisation requires approximately 10–12 GB of VRAM, exceeding current XR headset budgets. A pure cloud call on every frame introduces an unacceptable round-trip latency. The architecture resolves this tension with a **hybrid decoupled topology**.

- **Edge / on-device layer.** Capture, Transport, and the Proactive Gate execute close to or on the device, keeping the continuous monitoring loop latency-local and requiring only minimal compute.
- **Local server layer.** Fusion & Reasoning (Gemma 4 E4B) runs on a local GPU server and is invoked only when $D_t \geq 1$. Because cooking-critical transitions unfold over seconds, the infrequent round-trip fits comfortably within the latency budget. Operating on a local server rather than a cloud API preserves the deterministic, open-weight inference properties of Gemma 4 E4B.

The design is supported by multiple empirical results in the corpus. Em-Garde achieves 10–15 fps on a single GPU precisely because the large model is called only at startup [25]. YETI’s 1 FPS SSIM gate matches heavy frameworks on recall, demonstrating that the gate handles most detection work [8]. StreamRAG reduces generation latency by 27% through visual-textual context reuse [1]. EC-RAG demonstrates that unbounded streaming without windowing causes linear out-of-memory growth, motivating the bounded SES window design of K_2 [18]. MiniCPM-V 4.5 confirms that 4 B-scale models are viable for real-time egocentric reasoning [2].

5 Evaluation

5.1 Study Design

The evaluation will adopt a **mixed design**: memory tier is a between-subjects factor and intervention mode is a within-subjects factor. Each participant will be assigned to one tier group and will cook scrambled eggs twice (once in Reactive mode and once in Proactive mode), with the order counterbalanced across participants within each group.

- **Group 1 (Single-tier).** Participants will experience two conditions in sequence: Reactive + Single-tier (K_1 recipe text only, assistant silent until queried) and Proactive + Single-tier (K_1 only, gate fires without a user prompt). This group provides the baseline for both H1/H2 and H3.
- **Group 2 (Dual-tier).** Participants will experience Reactive + Dual-tier (K_1+K_2 fusion, fires only on a user query) and Proactive + Dual-tier (full system, Section 4.9). This group provides the H3 contrast against Group 1.

The within-subjects Reactive vs. Proactive comparison, available in both groups, will test H1 and H2. The between-subjects Single-tier vs. Dual-tier comparison on guidance relevance will test H3. Because every participant cooks the same recipe twice, a learning effect on Trial 2 is expected; strict counterbalancing is therefore enforced—half of each tier group will experience Proactive mode first, the other half Reactive first—so that natural user adaptation is distributed evenly across conditions and system utility can be isolated from familiarity gains.

Task. The cooking task will be to prepare one serving of scrambled eggs while wearing an XR headset in a controlled laboratory kitchen. This task is chosen because it exhibits the time-critical, non-uniform intra-object state transitions (liquid $\rightarrow$ coagulating $\rightarrow$ solid $\rightarrow$ scorched) that are the canonical test case for the architecture and that static final-outcome models systematically fail to capture [15, 23]. All participants will be tested in the same standardised kitchen environment with identical equipment (same pan model, induction hob, and egg source) to minimise confounds. The XR headset will be confirmed based on hardware availability at the time of execution.

Population. Participants will be adults with no professional cooking training who are comfortable wearing a head-mounted display. Individuals with severe visual or hearing impairment that would preclude use of the XR display or audio alert channel will be excluded.

5.2 Evaluation Metrics and Instruments

5.2.1 Dependent Variables

H1 – Interruption timing accuracy and procedural error rate. *Interruption timing accuracy* will be operationalised as $\delta t = |t_{\text{int}} - t^*|$, the absolute temporal distance between the system’s intervention timestamp and an annotator-defined optimal intervention window for each critical recipe step [19]. Session video logs will provide the timestamps; lower δt indicates better timing. *Procedural error rate* will be the count of unrecoverable errors per session (e.g., burnt eggs, skipped steps with lasting consequences) scored from blinded video review. A proactive policy that fires at the right moment (before the state transitions beyond recovery) is expected to reduce both δt and error rate compared with the reactive baseline [26].

H2 – Perceived intrusiveness. Intrusiveness will be measured with a 7-point Likert item adapted from the AR Cooking and Sensible Agent instruments: “*The assistant interrupted me at the wrong moment*” [4, 9]. Two supplementary items will capture helpfulness and annoyance. These items will be administered immediately after each condition block to reflect the specific experience, not a retrospective average. The prediction for H2 is that proactive interventions will not score higher on intrusiveness than reactive querying despite delivering alerts without prompting, because well-timed interruptions carry a lower split-attention cost than the mental context-switch required to formulate and issue a query [20].

H3 – Guidance relevance. Relevance will be measured on two levels. *Subjective relevance* will be a 7-point Likert rating collected per intervention (or per self-initiated query in reactive conditions) via a post-hoc video-replay protocol: participants will review each notification clip and rate “The suggestion matched what was happening at that moment.” *Objective correctness* will be scored by a blinded rater who classifies each delivered suggestion as correct-step, incorrect-step, or safety-critical; the proportion of correct + safety-critical suggestions per condition will serve as an objective proxy [15].

5.2.2 Shared Instruments

The **System Usability Scale (SUS)**, a validated 10-item, 5-point Likert instrument scored 0–100, will be administered once after both condition blocks to provide a standardised, literature-comparable usability score [4, 9].

An **abbreviated NASA Task Load Index (NASA-TLX)**, limited to the Mental Demand and Frustration subscales, will be collected after each condition block as a lightweight proxy for perceived cognitive load consistent with the split-attention model used throughout the architecture design [20].

A **short semi-structured interview** (approximately five to ten minutes) will close the session. Questions will cover: when the assistant helped or hindered the task; whether the participant felt in control; and whether they would use such a system while cooking unsupervised. Responses will be analysed thematically to triangulate the quantitative results and identify interaction patterns not captured by the Likert items [11].

5.3 Procedure

Each session will proceed as follows.

1. **Screening and consent** (≈ 10 min). Eligibility will be confirmed, informed consent obtained, and the XR headset fitted and calibrated to the participant.
2. **Familiarisation** (≈ 5 min). The participant will cook one practice serving of scrambled eggs with the system switched off. This step will establish baseline recipe familiarity and headset comfort; no data will be collected.
3. **Condition block 1**. The experimenter will silently configure the system to the participant's first assigned intervention mode. The participant will cook one serving of scrambled eggs with the system active; the session will be video-logged for post-hoc annotation. Immediately after, post-condition Likert items and the abbreviated NASA-TLX will be administered (≈ 5 min).
4. **Short break and kitchen reset** (≈ 5 min). The cooking station will be returned to its standardised starting state (fresh pan, fresh eggs).
5. **Condition block 2**. The procedure of step 3 is repeated under the second intervention mode.
6. **Post-session SUS**. The System Usability Scale will be completed once after both blocks (≈ 5 min).
7. **Semi-structured interview** (≈ 5 – 10 min).

The estimated total session duration is 45–60 minutes. The study protocol will be submitted for ethics review prior to data collection, in accordance with the open-science policies adopted by SIGSOFT (see <https://github.com/acmsigsoft/open-science-policies>).

5.4 Expected Outcomes and Limitations

The evaluation is designed to provide evidence on three fronts. First, a lower δt and fewer unrecoverable errors in Proactive conditions relative to Reactive conditions would support H1. Second, equal or lower intrusiveness scores in Proactive conditions would support H2, confirming that unprompted alerts are not perceived as more disruptive than user-initiated queries in a hands-busy task. Third, higher subjective and objective guidance relevance in Dual-tier groups relative to Single-tier groups would support H3, demonstrating that fusing live state tracking with static recipe knowledge produces meaningfully better suggestions.

Several limitations apply. The controlled kitchen setting maximises internal validity but may not generalise to the variability of real home cooking environments. The scrambled eggs task, while well-suited to the architecture's test case, represents a single, short procedure; longer or more complex recipes may reveal latency and memory-management behaviours not visible in a five-minute cook. Finally, the study will rely on participant ratings for guidance relevance, which are subject to response-format effects; the blinded-rater correctness proxy is included precisely to provide an independent check on the subjective data.

6 Work Schedule

The following Gantt chart (Figure 2) outlines the planned timeline and key stages of the proposed project.

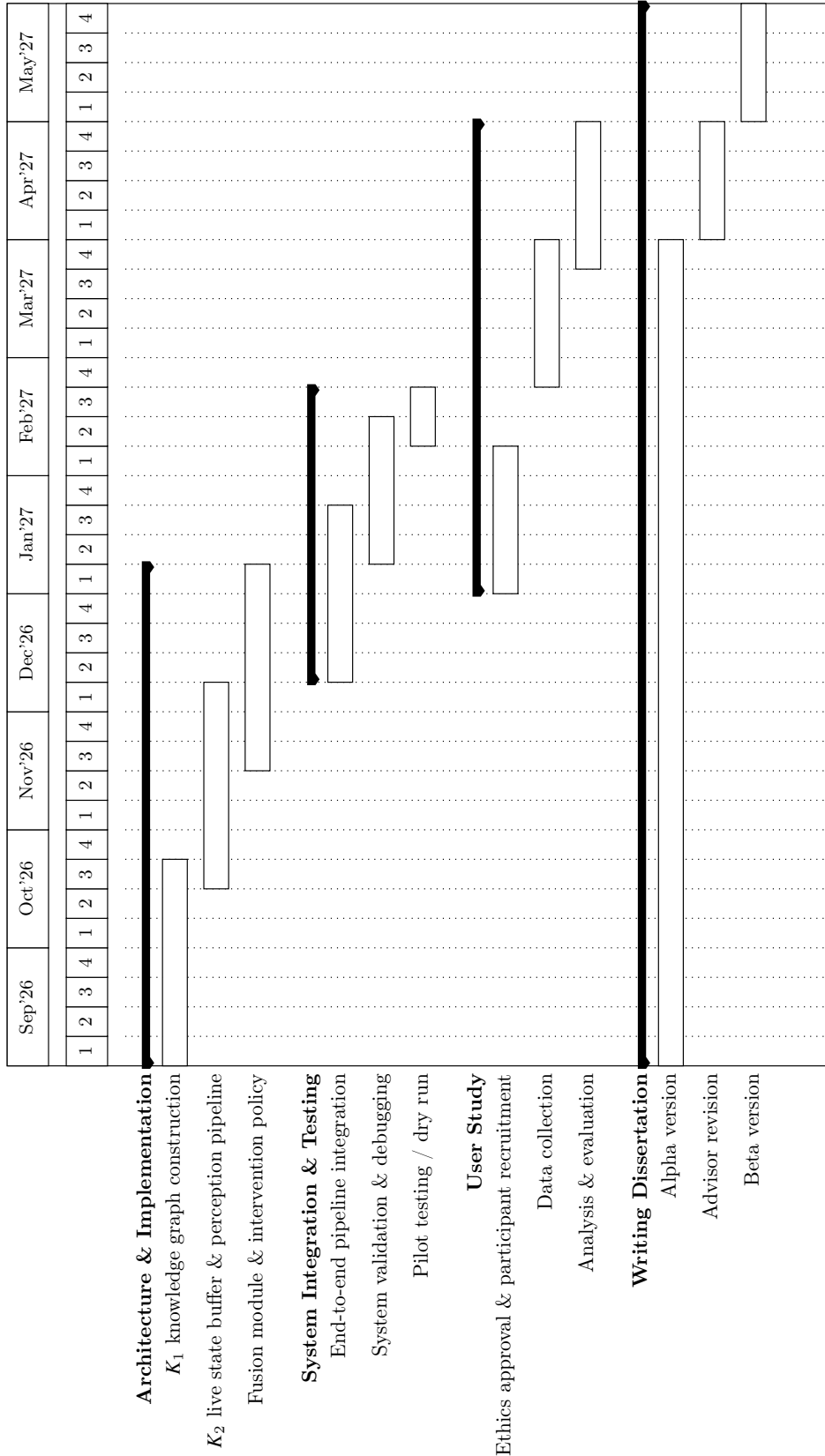


Figure 2: Planned Schedule

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